

Лабораторная работа 1 Кочетковой М.П.

2 курс. ИВТ-2.2

1234₁₀

3456₁₀

$$\begin{array}{r}
 1234 \div 2 = 617 \text{ remainder } 0 \\
 617 \div 2 = 308 \text{ remainder } 1 \\
 308 \div 2 = 154 \text{ remainder } 0 \\
 154 \div 2 = 77 \text{ remainder } 0 \\
 77 \div 2 = 38 \text{ remainder } 1 \\
 38 \div 2 = 19 \text{ remainder } 0 \\
 19 \div 2 = 9 \text{ remainder } 1 \\
 9 \div 2 = 4 \text{ remainder } 1 \\
 4 \div 2 = 2 \text{ remainder } 0 \\
 2 \div 2 = 1 \text{ remainder } 0
 \end{array}$$

$$1234_{10} = 10011010010_2$$

$$\begin{array}{r}
 3456 \div 2 = 1728 \\
 1728 \div 2 = 864 \\
 864 \div 2 = 432 \\
 432 \div 2 = 216 \\
 216 \div 2 = 108 \\
 108 \div 2 = 54 \\
 54 \div 2 = 27 \\
 27 \div 2 = 13 \text{ remainder } 1 \\
 13 \div 2 = 6 \text{ remainder } 1 \\
 6 \div 2 = 3 \text{ remainder } 0 \\
 3 \div 2 = 1 \text{ remainder } 1
 \end{array}$$

$$3456_{10} = 110110000000_2$$

$$\begin{array}{r}
 2) \quad 110110000000_2 \\
 \quad 1001100010_2 \\
 \hline
 1001001010010_2
 \end{array}$$

$$\begin{array}{r}
 3) \quad -3456_{10}: \quad 01101100000000_2 \\
 \quad \quad \quad 100100111111_2 \\
 \hline
 \quad \quad \quad 10010100000000
 \end{array}$$

$$\begin{array}{r}
 \quad \quad \quad 10010100000000_2 \\
 \quad \quad \quad 10011010010_2 \\
 \hline
 \quad \quad \quad 10111010010_2
 \end{array}$$

$$4) \begin{array}{r|l} 1 & 16 \\ 2 & 7 \\ 3 & 7 \\ 4 & 4 \\ \hline & D \end{array}$$

$$1234_{10} = 4D2_{16}$$

$$\begin{array}{r|l} 3 & 16 \\ 4 & 2 \\ 5 & 1 \\ 6 & 0 \\ \hline & 8 \end{array} \quad \begin{array}{r|l} 16 \\ 1 & 16 \\ 3 & 1 \\ \hline & D \end{array}$$

$$3456_{10} = D80_{16}$$

$$5) \begin{array}{r} 4D2_{16} \\ D80_{16} \\ \hline 1252_{16} \end{array}$$

$$6) \begin{array}{r} -D80_{16} \\ 4D2_{16} \\ \hline 8AE_{16} \end{array}$$